

Project Initialization and Planning Phase

Date	30 September 2026
Team ID	xxxxxxx
Project Title	AI StudyBuddy - AI-Powered Study Material Assistant for Students
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform the way students study using generative AI, boosting learning efficiency and exam readiness. It tackles the slow, manual effort of preparing revision material, promising faster revision, better self-testing, and more confident students. Key features include AI-generated summaries, flashcards, quizzes and a personalised study plan.

Project Overview	
Objective	The primary objective is to help students turn their own study material into a summary, flashcards, a quiz and a study plan using generative AI, making revision faster and more effective.
Scope	A web application with a Node.js/Express backend and MongoDB database, covering secure user login, file upload, AI content generation and storage of results for each user.
Problem Statement	
Description	Students spend a lot of time manually condensing long notes, writing practice questions and planning study schedules, which is slow and inconsistent and often leaves them unprepared for exams.
Impact	Solving this will save study time, improve revision quality and self-testing, reduce exam stress, and help students stay organised and confident in their preparation.
Proposed Solution	
Approach	Using a generative AI model (Google Gemini) through an Express API to read uploaded material and produce a summary, flashcards, a quiz and a study plan, all stored per user so they can be revisited anytime.
Key Features	- Secure registration and login using JWT tokens and bcrypt password hashing. - Role-based access control for students and admins. - Upload of study material (.txt, .md, .pdf) up to 5 MB. - AI-generated summary, flashcards, quiz and study plan using Google

	Gemini. - Automatic retry when the AI service is temporarily unavailable. - Every user's materials and generated content saved in MongoDB for later revision.
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	2 vCPU cloud server or laptop (no GPU needed, AI runs on Gemini API)
Memory	RAM specifications	4 GB
Storage	Disk space for data, models, and logs	20 GB SSD
Software		
Frameworks	Node.js frameworks	Express.js
Libraries	Additional libraries	mongoose, bcryptjs, jsonwebtoken, multer, cors, dotenv, express-async-errors, @google/generative-ai
Database	Data storage	MongoDB
AI Model	Generative AI API	Google Gemini (gemini-3.7-flash)
Development Environment	IDE	VS Code, Node.js
Data		
Data	Source, size, format	User-uploaded study material, max 5 MB per file, .txt / .md / .pdf